



Trade credit, payment duration, and SMEs' growth in the European Union

Vivien Lefebvre¹

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Abstract

Long payment durations are commonly considered a major drag on small- and medium-sized enterprises (SMEs) growth by public policies and practitioners. At the same time, long payment durations mean that customers have more time to assess to product quality, which reduces information asymmetries and thus facilitates the establishment of strong business relationships. Building on these conflicting views, we investigate the relationship between payment duration and two dimensions of firm growth, namely growth in sales and growth in employment. Our results, based on a large sample of SMEs located in the European Union, show a positive relationship between payment duration and growth. Moreover, we observe that in countries characterized by a strong quality of payment conditions, the positive relationship between payment duration and growth is amplified. We interpret our findings as evidence that SMEs, by accepting long payment durations, act positively upon their ecosystem by facilitating other SMEs financing conditions, thus creating a “virtuous cycle” in environments characterized by supportive institutional context.

Keywords Trade credit · SMEs · Institutional context

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✉ Vivien Lefebvre
vivien.lefebvre@em-strasbourg.eu

¹ LaRGE Research Center - EM Strasbourg Business School, University of Strasbourg, 61 Avenue de la Forêt Noire, Strasbourg 67000, France

Introduction

Small- and medium-sized enterprises (SMEs) represent the core of the economy in terms of employment and value added. In the European Union (EU), SMEs account for more than half of the value added and two thirds of jobs (European Commission, 2020). The major role played by SMEs explains why many governments consider SMEs' growth as a major issue. Over the last twenty years, public policies were designed to sustain SMEs' development and entrepreneurship (Chowdury et al., 2019). A critical issue in SMEs' development is undoubtedly related to payment duration in business transactions. For instance, late payment affects about 80% of European firms (European Commission, 2015) and is considered a major hurdle to job creations by one out of five companies in the EU. On average, firms lose more than 2% of their annual revenues because of bad debt in the EU (Intrum Justitia, 2019). Notwithstanding the problem of late payment, in this paper, we take an opposing view and propose that there are also benefits to the issuance of trade credit for SMEs growth.

Despite its important economic impact, payment duration and its consequences on SMEs growth have received modest academic attention, especially outside the UK context. Howorth and Wilson (1998) and Pike and Cheng (2001) document the heterogeneity of credit management practices and their impact on firm's financial management and performance in the case of UK firms. Peel, Wilson, and Howorth (2001) show that SMEs' managers have strong expectations about public policies designed at limiting late payment. Interestingly, about ten years later Paul and Boden (2011) observe that current regulation and management of legal payment duration terms remain inadequate in the UK context. More recently, Martínez-Sola, García-Teruel, and Martínez-Solano (2013) show that granting more trade credit is a promising strategy for financially unconstrained Spanish SMEs to increase profitability. Hill, Kelly, and Lockhart (2012) make a similar observation in the case of large publicly listed firms in the US. With respect to institutional factors, Barrot and Nanda (2020) document the effect of the Quickpay reform in the USA through which the government accelerated payments to small businesses and led to employment growth. However, what influence the institutional context has on trade credit between firms as a channel to growth remains an open question.

Our goal in this paper is (1) to assess the benefits of granting trade credit for SMEs growth and (2) to investigate the role played by the quality of payment conditions at the country-level on the value of trade credit granted by SMEs on growth in an attempt to extend our knowledge on the influence of institutions on entrepreneurial activity (Chowdury et al., 2019). Although a great amount of work has investigated the role of trade credit as an alternative to banking debt to finance SMEs (see for instance Casey & O'Toole 2014; Gama & Van Auker, 2015; McGuinness & Hogan, 2016), little is known about the value of granting trade credit to sustain SMEs' growth. Granting trade credit is a way to foster strong business relationships with other firms by building trust and reducing information asymmetries between suppliers and customers (Smith, 1987; Long et al., 1993; Ng et al., 1999; Martínez-Sola et al., 2013).

However, the internal resource constraints faced by SMEs condition their ability to grant trade credit (Wilson & Summers, 2002). Furthermore, granting trade credit

may not be a choice for SMEs but simply reflect the bargaining power larger customers have (Wilson & Summers, 2002; Wu et al., 2020). In this case, it seems less likely that longer payment duration provides any benefits to SMEs. The practitioners' view on the role of trade credit granted by SMEs is rather that it increases the risk of non-payment, makes cash-flows uncertain, and thus negatively affects growth (Intrum Justitia, 2019; European Commission, 2015). These opposed views contribute to motivate our study to gain empirical insights on the benefits (or limitations) of payment duration on growth and the influence of the institutional context, since such information would be useful both for policy makers and managers.

We conduct an international study in EU countries to take advantage of the Late Payment Directive (LPD) adopted in 2011 in the EU and integrated in national laws in 2013. The LPD aimed at reducing the payment duration of public authorities to thirty days and the payment duration of firms to sixty days. The LPD also established penalties and interest for late payment¹. Interestingly, the impact of the LPD across EU countries is highly heterogeneous in terms of observed payment delays and perceived business conditions by EU SMEs (Intrum Justitia, 2019; European Commission, 2016). We use Bureau Van Dijk's Amadeus database to collect financial and accounting data on 104,952 European SMEs over the period 2010–2018. Our sample of SMEs is thus large and allows to distinguish empirically between different categories of SMEs (micro, small, and medium firms). Focusing on the 2010–2018 period means that we observe SMEs' growth and trade credit before, during, and after the implementation of the LPD and thus ensures variability in the institutional context. To measure the trade credit conditions at the country-level, we use the European Payment Risk Index (EPRI) developed by Intrum Justitia, a well-known credit manager in Europe. The EPRI assesses the strength and quality of legal payments conditions and is accessible in annual reports published by Intrum Justitia. Our empirical analysis is based on regression analysis and we use a propensity score-matching algorithm as a robustness test to ensure the credibility of our findings.

Our results show a positive relationship between payment duration and growth, suggesting that the benefits of longer payment duration offset the costs of granting trade credit for SMEs. This positive relationship is magnified in countries where the strength and quality of legal payment conditions are high. Interestingly, we document that the benefits of payment duration and legal context hold for growth in sales and growth in employment, highlighting the strong implications of this study for public policies and managers.

Theoretical framework and hypotheses development

Use of trade credit and growth

In most business transactions, products and services are sold on credit rather than cash, which creates accounts receivables and represents trade credit. The amount of trade credit (payment duration) granted to customers is a key characteristic of credit

¹ https://ec.europa.eu/growth/smes/sme-strategy/late-payment_en.

terms that is influenced by industry-standards, market power, financial constraints, and strategic choices among other factors (Mian & Smith, 1992; Ng et al., 1999; Wu et al., 2020). There are several theoretical factors that explain why firms (suppliers) extend trade credit to other firms (customers)².

First, imperfect information in business transactions leads to transactions costs. When payment is made cash, the customer is exposed to poor product quality because he/she does not have time to assess product quality (Smith, 1987; Lee & Stowe, 1993). When payment is made on credit, the supplier is exposed to credit risk, especially if customers are aware of their poor creditworthiness (Ng et al., 1999). Thus, both parties need devices that reduce information asymmetries in order to limit transaction costs. Reputation and legal context represent two examples of such devices. Interestingly, the idea that granting longer payment terms to customers is a way to increase sales by reducing information asymmetries about product quality finds strong ground for small firms (Long et al., 1993), which often do not have a strong established reputation (Stinchcombe, 1965; Aldrich & Auster, 1986). It follows that granting trade credit is a channel through which SMEs can increase sales and grow. Of course, it is important to acknowledge that such a growth strategy is costly, since granting trade credit increases working capital requirements, which are often financed through banking debt (Chen & Kieschnick, 2018).

At the same time, granting trade credit may not be a choice for “small” suppliers. Firms that have a strong market power can impose payment terms to maximize the availability of internal financial resources and decide whether and to whom they delay payment (Wu et al., 2020). Note that the market power argument suggests that customers that have a strong market power can set initial credit terms that best suit them, but also that these firms may decide to pay after the payment duration initially set in business contracts (late payment). Because SMEs often have a very limited market power, the extension of trade credit to customers may simply reflect their limited bargaining power toward large customers. In this case, longer payment duration indicates that SMEs make efforts to adjust to large customers’ requirements.

A second theoretical argument that motivates the use of trade credit to increase sales is the discrimination customers can operate when multiple suppliers offer substitutable products or services. Assuming product or service quality is comparable across suppliers, customers likely opt for the supplier that offers the best payment conditions, and in particular the longest payment duration (Shipley & Davies, 1991). For financially constrained customers, long payment duration via trade credit is an important substitute to banking debt (Petersen & Rajan, 1997). In the particular case of SMEs, it may be difficult for SMEs’ managers to identify and seize the opportunity to grant trade credit to financially constrained customers. Indeed, financial management is notably poor among SMEs because SMEs’ managers lack time for reviewing and adjusting trade credit practices (Peel & Wilson, 1998; Peel et al., 2001; Lindeloef & Løefsten, 2005). Furthermore, SMEs acting as potential suppliers of trade credit

² We acknowledge that trade credit is also generated in the case of pre-payments, which occur when customers extend trade credit to suppliers. As described by the financial bootstrapping literature, pre-payments represent an alternative source of finance for young and fast-growing firms (see for instance Winborg & Landström 2001). However, we consider the study of pre-payments to be outside the scope of this paper.

are themselves likely financially constrained, making the issuance of trade credit extremely costly.

However, our point is not to propose that SMEs are more or less likely to grant trade credit and long payment duration but rather to stress the benefits (and limitations) of doing so. As shown by Brinckmann, Salomo, and Gemuenden (2011), strategic financial management is a powerful tool for young and small firms to grow but requires the presence of specialized financial managers with adequate skills. Recent papers have highlighted that SMEs face significant opportunity costs when they fail to adjust their financial management and trade credit practices. For instance, sales can be lost when payment duration is too short, especially for financially constrained firms (Baños-Caballero et al., 2012; Lefebvre, 2021, 2022).

To put it shortly, all firms likely benefit from extending trade credit because accepting longer payment duration facilitates growth through stronger business relationship and reduces the risk of opportunity costs resulting from lost sales. These benefits are, however, stronger for SMEs whose reputation is less developed than that of large firms because this lower reputation increases information asymmetries for customers:

Hypothesis 1 *There is a positive relationship between payment duration (extension of trade credit) and growth for SMEs.*

Legal payment conditions and the relationship between trade credit and growth

Public policies cause important institutional changes that have powerful consequences for firms' managers (Scott, 2008). Not only do institutional changes shape business environment but they also force managers to take these changes into account (North, 1990) and act as facilitators or "external enablers" of entrepreneurial activity (Kimjeon & Davidsson, 2022). The impact of legal reforms affecting business transactions is a core topic in institutional theory and at the nexus of institutional theory and entrepreneurship (see for instance Eberhart et al., 2017). Still, we know of no previous studies exploring the role of payment conditions enforcement laws on firm growth. Our core argument is relatively intuitive. Although it takes time for institutional changes in payment conditions to spread across the economy (European Commission, 2016), these changes contribute to strengthen the late payment laws enforcement and speed up cash collection periods. Both factors reduce uncertainty and risk in business transactions and hereby facilitate SMEs growth (Scott, 2008).

Over the past decade, public policies and practitioners have stressed the need to reduce payment duration and avoid late payment to strengthen SMEs' financial health and growth ability. In 2011, the EU adopted the LPD to protect SMEs from the negative impact of late payment in the aftermath of the Global Financial Crisis. The major goals of this directive were to reduce payment duration to thirty days for public-to-private transactions and sixty days for business-to-business transactions, and to establish automatic penalties and a reference interest for late payment. The LPD was transposed into national legislations in 2013, but its effects on payment conditions appeared limited in the following years (European Commission, 2015). Payment duration slowly decreased in the EU following the LPD but the greatest

progresses were observed in the private sector, while public authorities fail to respect payment limits.

Interestingly, late payment in private sector is highly correlated with late payment in public sector for EU countries (European Commission, 2016). Despite the LPD, few firms, especially among the smallest SMEs, exercise their recovering and compensation rights when faced with late payment, mostly out of fear of losing customers. Improving payment delays is a major concern for all countries and not only the EU. In the USA, the Quickpay reform aimed at reducing the payment delays faced by small firms that worked with the government in an attempt to reduce financial constraints for small firms. The Quickpay reform led to benefits in terms of employment growth but with important differences depending on the labor market in which firms were active (Nanda and Barrot, 2020).

As we previously discussed, extending trade credit to customers, even for SMEs, is a common business practice and is not intrinsically a factor that harms firms' growth. However, extending trade credit requires that firms spend time in the credit-management process, assess customers' credit risk, set or discuss credit terms, and finance trade credit until payment is made (Mian & Smith, 1992). SMEs often do not have the time and skills to perform these various tasks (Peel & Wilson, 1998; Peel et al., 2001). As a consequence, SMEs are highly vulnerable to "bad operating debt", meaning that their lower ability to assess credit risk increases the likelihood that they accept unreliable customers, that eventually do not pay for the products and services they purchase. This problem is even more serious because unlike larger firms, SMEs depend extensively on internally generated money to finance their activities (Carpenter & Petersen, 2002). Indeed, the high degree of information asymmetry that characterizes SMEs constrains their ability to borrow money externally (Stiglitz & Weiss, 1981; Berger & Udell, 1998; Cassar, 2004).

It follows that the adoption of stricter payment duration enforcement laws, like the LPD, should increase the benefits of granting trade credit. First, by reducing the duration of legal payment delays, the LPD creates a homogenous framework across the EU that partly rebalances the disparity in bargaining power between SMEs and large firms. Second, the diffusion of information related to trade credit practices and laws gives SMEs' managers additional knowledge and enforcement tools about credit management. These factors improve SMEs' managers understanding of the strategic role of trade credit and increase SMEs' managers ability to use it. SMEs' managers can thus more easily decide to use (or not) trade credit extension and do so within an institutional context characterized by stronger enforcement laws that reduce uncertainty and risk of late payment. We thus propose that:

Hypothesis 2 *Stronger legal payment conditions positively moderate the relationship between payment duration and growth for SMEs.*

We summarize our hypotheses with the following conceptual model:

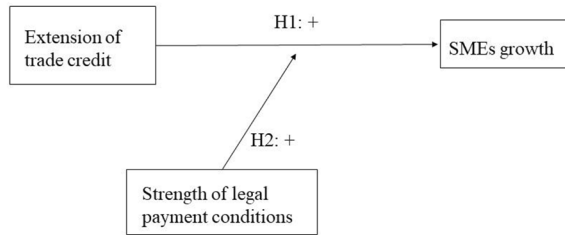


Fig. 1 Conceptual model and hypotheses

Methodology

Sample and data

We use three different sources to collect our data. Financial and accounting information about European SMEs come from the Amadeus database provided by Bureau van Dijk. Amadeus is a widely used source of data in finance and management studies. However, it is important to notice that Amadeus' coverage depends on national reporting requirements, making its coverage heterogenous across countries. Thus, we exclude countries for which Amadeus coverage is poor³ due to poor national reporting requirements. Specifically, we exclude Ireland, UK, Denmark, and Luxembourg (Kalemli-Özcan et al., 2015). Information about payment duration and risk in European countries comes from the European Payment Reports of Intrum Justitia, an important credit management company. Specifically, we use the EPRI displayed by Intrum Justitia throughout the years 2010 to 2017⁴. Last, we make use of the World Bank national economic information to collect country-specific variables.

We collect our information over the 2010–2018 period because one year of data is lost to calculate the growth variables. We exclude publicly listed firms and firms operating in the finance industry. Indeed, publicly listed firms have a better access to external finance and a more developed reputation than privately held firms. These factors typically affect trade credit practices as well as growth speed. Furthermore, payment duration in the finance industry is out of the scope of this paper. At this stage, the sample is made of 393,896 firms that represent 2,471,142 firm-year observations. Since we focus on SMEs, we exclude firms that do not meet the European Commission's definition of SMEs (European Commission, 2020). Thus, firms are considered SMEs if (1) they do not belong to a business group, (2) they have less than 250 employees, (3) either their sales turnover is lower than 50 M€ or their total assets

³ We provide in the appendix the number of observations for each country. As can be observed, Italy and Spain are largely represented while we only have a few dozens or a few hundred observations for some others. Other studies that used Amadeus to create international datasets typically suffer from the same limitation (see for instance Vanacker et al. (2017)).

⁴ Unfortunately, Intrum ceased the publication of the EPRI in its European Payment Reports after 2017. This constrains our dataset. The EPRI covers most countries in the European Union with the notable exceptions of Luxembourg.

is lower than 43 M€. Firms affiliated to a BG commonly have important business transactions with other firms affiliated to the same BG. It has been documented that common financial management within BG is performed via the trade credit channel to provide assistance to the most vulnerable affiliated firms (Deloof & Jegers, 1996, 1999). It is thus important to exclude business group affiliated firms in our study. With respect to business group affiliation, Amadeus directly provides information about the number of entities within the business group to which a firm is affiliated. This number is zero if a firm stands alone. To correctly identify SMEs, we also need information about the number of employees in each firm. Because this information is often missing, we can only obtain information for 104,952 SMEs that represent 605,212 firm-year observations.

Variables

Dependent variables

The focus of the paper is the relationship between payment duration (trade credit extended to customers) and growth. Measuring growth has proved to be a controversial issue because academics, policy makers, and entrepreneurs have different views about the dimension of growth that can be considered a desirable outcome (Achtenhagen et al., 2011). We therefore use two measures of growth in all regressions: the annual percentage change in sales and the annual percentage change in the number of employees. Doing so ensures that our results speak both to entrepreneurs and managers, who view growth in sales as a variable of interest, and to policy makers, who pay more attention to growth in employment. Interestingly, previous research reported that these two dimensions of growth are poorly correlated (Shepherd & Wiklund, 2009). As will be seen, the results obtained in the present study do not differ much depending on whether sales growth or employment growth is considered.

Independent variables

We use two independent variables. The first independent variable is payment duration. We measure payment duration as accounts receivables divided by sales, which is a common practice in the literature on trade credit (Martínez-Sola et al., 2013; McGuinness & Hogan, 2016). Of course, this measure is relatively crude, for instance because it does not take into consideration industry- and country-characteristics, neither does it account for endogenously determined firm-specific factors. Therefore, we discuss two alternative measures of payment duration (annual mean-industry adjusted accounts receivable on sales and a regression-based measure of payment duration) as robustness tests and obtain comparable results.

The second independent variable is the EPRI provided by Intrum Justitia, which is a measure of a country's global risk of late payment and the impact of late payment for this country's firms calculated on an annual basis⁵. The EPRI takes the

⁵ The index is based on the effective payment duration reported by firms, the age structure of receivables, the fraction of sales lost because of bad operating debt, an estimation of country-specific risk trends, the

form of an index that ranges between -2 (poor payment stability, high risk) and $+2$ (high payment stability, very low risk). The EPRI is calculated by Intrum Justitia via annual waves of survey sent to a large number of firms across the EU. Figure 2 below displays the EPRI in 2010 and in 2017 for each country in our sample. As can be seen, there are heterogenous levels and changes in EPRI across our sample countries. Sweden (change in EPRI over the period is -0.31) and Finland (-0.24) are countries with strong EPRI scores but see their EPRI decrease throughout the considered period, despite the LPD. France ($+0.12$), Germany ($+0.09$), Belgium ($+0.01$) and the Netherlands ($+0.15$) are countries with relatively stable and average EPRI. Greece (-0.06), Italy (-0.30), and Portugal (-0.38) are countries with low and decreasing EPRI, which reflects poor and deteriorating payment conditions. Last, and more interestingly, Austria ($+0.55$) and Spain ($+0.30$) are the countries where the EPRI increase was the strongest, suggesting that these two countries are also those where the LPD was more actively transposed into national payment conditions laws. It is important to notice that the starting point is quite different for Austria and Spain. The EPRI of Austria in 2010 is comparable to that of Germany and among the “good students” of the EU. Spain, on the opposite, has one the lowest EPRI of all the considered countries in 2010.

Control variables

We include several time-varying control variables in our estimations that are known to influence growth. First, we use firm-level controls. We control for firm size, mea-

characteristics of the consequences of late payment in terms of national enforcement procedures, and the causes of late payment among other factors. It is worth noting that the EPRI was calculated as a score between 100 and 170 points until 2014. We apply a linear transformation to convert the 2010–2013 EPRI into a $[-2; +2]$ index.

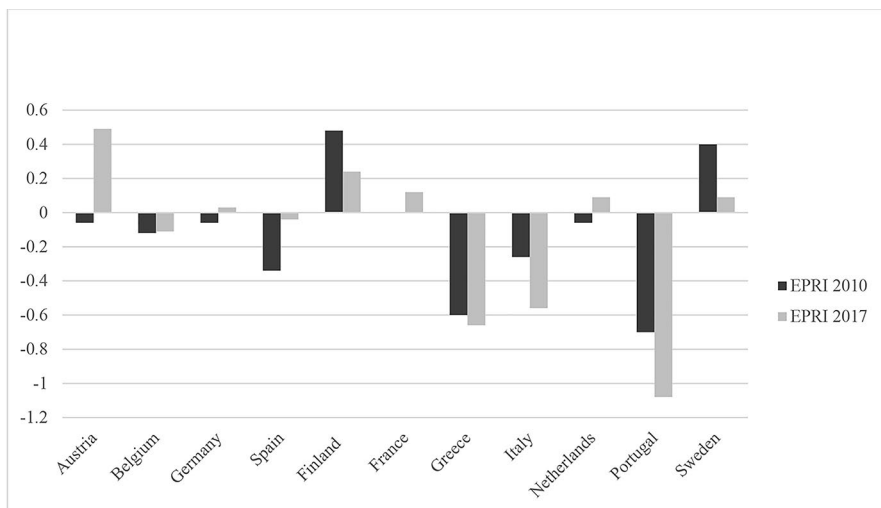


Fig. 2 Change in the European Payment Risk Index over the 2010–2017 period

sured as the natural logarithm of total assets, because smaller firms grow faster than larger firms (Almus, 2000; Calvo, 2006). Note that we do not use sales or number of employees as measures of size because sales and number of employees are used to calculate the dependent variable. We also control for cash holdings because cash represents organizational slack that can enhance growth by facilitating the exploration of new business opportunities but also harms growth by inducing entrepreneurial laziness (Bradley et al., 2011). Cash holdings are measured as cash and equivalents divided by total assets. We control for the degree of indebtedness, measured as financial leverage (short term debt plus long term debt divided by total assets) because access to debt is a critical factor of SMEs growth and survival (Berger & Udell, 1998; Honjo & Harada, 2006). We also include a control for profitability, because more profitable firms have a greater ability to retain profits which can facilitate future investments although empirical evidence on this question is mixed (Bottazzi et al., 2008). Profitability is measured as return-on-assets (ROA) calculated as operating income divided by total assets.

Second, we include industry-level controls because trade credit practices and growth speed largely depend on industry specificities (Mazzucato & Parris, 2015; Wu et al., 2020). All industry-based calculations use the Nomenclature statistique des Activités économiques dans la Communauté Européenne (NACE) codes at the four-digit level. NACE codes are the European equivalent of the commonly used Standard Industry Classification (SIC) codes. The use of the four-digit level ensures a fine-grained control of industry characteristics. We control for the annual average growth rate (either in sales or in employment depending on the dependent variable considered) by industry and country to take into account that national public policies may be designed to foster growth in specific industries. We also control for the annual average size (based on total assets) of competitors within each industry and country, because the presence of many large firms implies that SMEs bargaining power is strongly limited. The last industry-level control is the Hirschmann-Herfindahl index of industry competition, which is calculated as the sum of the squared market share of each firm within an industry for each country. Indeed, strong industry competition increases the risk of bankruptcy (Utterback & Suárez, 1993).

Third, we include controls at the country-level, namely the natural logarithm of GDP per capita and the rule of law score. GDP per capita controls for the degree of economic development and rule of law controls for the degree of law enforcement. Control for rule of law is especially important in our study, because one could argue that when rule of law is high in a given country, then the likelihood of late payment is lower. To ensure that our EPRI measure does not simply reflect rule of law, it is important to include explicitly rule of law in our model (see for instance Vanacker, Collewaert, and Zahra (2017) for a similar approach). Both GDP per capita and rule of law are time-variant measures taken from the World Bank. We winsorize all our variables at the first and last percentiles to limit the influence of outliers.

Econometric approach and endogeneity concerns

Our goal is to study the relationship between payment duration, institutional payment-risk, and firm growth. We thus decide to use panel data gathered at the firm-level. The

use of panel data facilitates control for unobserved time-invariant heterogeneity at the firm-level via firm-fixed effects regressions and further ensures that remaining time-invariant heterogeneity at the industry- and country-level are taken into account. To ensure that fixed-effects is the correct specification, we run a Hausman test. The chi-squared statistics is 3,193.35 and the corresponding p-value is lower than 0.000, so we reject the null hypothesis that the errors are not correlated the regressors, which confirms that a fixed effects model is the correct specification. Our main estimation throughout the paper is therefore a firm- and year-fixed effects model. The inclusion of year-fixed effects means that other macroeconomic factors do not drive our results.

The model we use limits the risk of omitted variable bias, but it is important to take into account other sources of endogeneity. Sample selection bias is a concern because one could argue that our results are driven by the fact that only surviving firms are observed and that these firms are typically those which cope with long payment duration. We thus check the proportion of firms that their sales drop to zero, indicating business failure. The proportion of business failure is always higher than 1% and reaches 3% in 2011 in our sample, which is rather in line with observed statistics in the EU context. Indeed, there has been a strong decline in business failures in the EU over the considered period⁶. It seems that sample selection bias is thus not a serious concern in our case.

With respect to reverse causality, we acknowledge that sales growth may drive payment duration. Typically, it is possible that firms that have high growth potential are more likely to accept longer payment duration in countries with lower payment risk. However, it seems less likely that employment growth drives payment duration, because the decision to hire employees is typically a long-term decision for SMEs while payment duration policy is a matter of short-term financial management. Additionally, whether SMEs can adjust payment duration is not straightforward, because, as we already discussed, their bargaining power with customers is often limited. To be clear, we do not claim to establish a clear causality channel between payment duration and growth in this paper. Still, we lag our independent and control variables by one year with respect to the dependent variable to attenuate the reverse-causality issue. We also present a formal test we conducted via propensity-score matching to mitigate reverse causality concerns in the robustness tests section.

Results

Univariate analysis

We begin our empirical analysis with descriptive statistics about the variables used in Table 1 below. The mean firm holds 0.975 M€ of total assets (median is 0.408 M€), which means that our sample mostly includes very small firms. The mean sales growth rate is 8.1% (median is 1.6%) and the mean employment growth rate is 5.7%

⁶ Declaration of bankruptcies rate drop by 20% between 2014 and 2016 in the EU (Source: Eurostat, accessible at https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Quarterly_registrations_of_new_businesses_and_declarations_of_bankruptcies_-_statistics&stable=0&redirect=no).

(median is 0%). The differences between mean and median growth rates suggest that a small number of firms represent most of jobs and value creation, an observation that is fully in line with the literature on firm growth (see for instance Coad et al., 2014). The mean firm holds 13.0% of its total assets as cash, and financial debts account for 16.7% of total assets. These figures indicate that our sample are not highly indebted toward banks. We present as an appendix some additional statistics at the country level.

Next, we present a correlation matrix in Table 2 below. Sales growth and employment are positively correlated with payment duration but negatively correlated country-level variables such as the EPRI and GPD per capita. We calculate the variance inflation factors (VIF) to ensure that multicollinearity, especially between country-level variables, does not affect our results. The highest VIF is 2.53 (EPRI) and the mean VIF is 1.44. These figures are well below the commonly accepted threshold of 5 above which multicollinearity is an issue.

Table 1 Descriptive statistics

Variables	Mean	Standard deviation	Minimum	25th percentile	Median	75th percentile	Maximum
Sales growth	0.081	0.525	-0.995	-0.087	0.016	0.134	4.862
Employee growth	0.053	0.331	-0.667	0.000	0.000	0.077	2.000
Size	6.882	1.313	3.258	6.014	6.880	7.734	12.231
Cash holdings	0.130	0.160	0.000	0.015	0.066	0.186	0.797
Leverage	0.167	0.198	0.000	0.000	0.089	0.284	0.853
Profitability (ROA)	0.051	0.107	-0.423	0.012	0.038	0.083	0.486
Asset tangibility	0.767	0.303	0.000	0.616	0.925	0.997	1.000
Industry sales growth	0.110	0.108	-0.080	0.050	0.088	0.146	0.474
Industry employment growth	0.064	0.059	-0.055	0.034	0.059	0.087	0.249
Size of competitors	7.032	0.784	5.364	6.552	7.019	7.446	9.113
Industry competition	0.081	0.120	0.002	0.016	0.039	0.098	0.672
Rule of law	1.341	0.317	0.084	1.066	1.138	1.650	2.096
GDP per capita	10.407	0.175	10.155	10.291	10.386	10.479	11.015
European Payment Risk Index (EPRI)	-0.476	0.347	-1.150	-0.820	-0.380	-0.280	0.440
Payment duration	0.240	0.245	0.000	0.032	0.183	0.352	0.968

N=605,212 for all variables except for employee growth where N=583,237 and industry employment growth where N=601,556.

Table 2 Correlation matrix

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 Sales growth														
2 Employee growth	0.161													
3 Size	-0.017	-0.044												
4 Cash holdings	-0.020	0.019	-0.197											
5 Leverage	0.009	-0.013	0.183	-0.315										
6 Profitability (ROA)	-0.060	0.062	-0.050	0.260	-0.132									
7 Asset tangibility	-0.023	-0.005	0.063	-0.028	0.060	0.038								
8 Industry sales growth	0.063	0.040	-0.052	0.004	-0.029	0.050	-0.025							
9 Industry employment growth	0.022	0.048	-0.120	0.034	-0.051	0.059	-0.033	0.465						
10 Size of competitors	0.007	-0.022	0.495	-0.092	0.144	-0.049	0.084	-0.104	-0.225					
11 Industry competition	0.015	-0.001	0.021	0.038	0.007	-0.005	-0.016	0.046	-0.029	0.123				
12 Rule of law	0.003	-0.017	0.169	0.049	0.191	-0.040	0.019	-0.167	-0.226	0.407	0.146			
13 GDP per capita	-0.007	-0.008	-0.144	0.055	-0.051	0.037	-0.006	0.209	0.069	-0.173	0.068	-0.139		
14 EPRI	-0.011	-0.016	-0.025	0.023	0.083	-0.004	-0.001	0.085	-0.060	0.031	0.087	0.248	0.653	
15 Payment duration	0.089	0.002	0.151	-0.189	0.069	-0.080	-0.014	0.034	-0.016	0.056	0.008	-0.080	-0.071	-0.043

N = 581,572. Correlation coefficients greater than 0.003 in absolute value are statistically significant at the 0.05 threshold.

Multivariate analysis

Main results

We run a preliminary set of fixed-effects regressions to better estimate the correlations between our variables. In Table 3 below, we present the results of firm- and year-fixed effects regressions obtained with the full sample of SMEs. Sales growth and employment growth are alternatively used as dependent variables. Payment duration and the EPRI are the independent variables, and we also include the interaction terms between the EPRI and payment duration. Several results are worth noting. First, we observe a positive and statistically significant relationship between payment duration and growth for both dimensions of growth. In other words, longer payment duration is associated with faster growth. Second, we observe a positive and statistically significant relationship between the EPRI and growth, indicating that firms located in countries with high EPRI grow faster than those located in countries with low EPRI. This observation is consistent with the idea that late payment risk is a factor that negatively affects growth. Third, the interaction term between payment duration and the EPRI is positive and statistically significant. In other words, the positive relationship between payment duration and growth is amplified in countries where the risk of late payment is low. These preliminary results are in line with hypotheses 1 and 2. Payment duration appears to be a channel to growth and not a constraint, and high payment conditions quality at the country-level amplify this channel.

It seems important to acknowledge that the statistical significance of our results may be related to the large size of the sample we use. Statistically significant results are not necessarily economically meaningful. A one-standard deviation increase in payment duration is associated with a 14.06% increase in sales growth and a 1.81% increase in employment growth. These figures suggest that payment duration has a relatively important economic impact on growth.

We refine the previous analysis in Table 4 below and present the results obtained for medium, small, and micro firms separately. Indeed, SMEs do not represent a homogenous category, especially with respect to financing decisions (Masiak et al., 2019). Firms are considered micro if they have less than 10 employees and either less than 2 M€ of sales or 2 M€ of total assets. Firms are considered small firms if they are not micro and if they have less than 50 employees and either less than 10 M€ of sales or less than 10 M€ of total assets. Firms are considered medium if they are not micro nor small and if they have less than 250 employees and either less than 50 M€ of sales or 43 M€ of total assets. Payment duration is positively and statistically significantly related to growth for all categories of firms. The interaction term between payment duration and EPRI is positive and statistically significant for all categories of firms and both dimensions of growth except in the case of employment growth for medium firms.

Interestingly, we observe that the coefficients of the payment duration term are larger for medium firms than for small or micro firms for both growth variables. The payment duration term for medium firms is statistically significantly higher than for small and micro firms ($p < 0.01$ for sales growth, $p < 0.1$ for employment growth) when we calculate the difference between these coefficients scaled by the squared

Table 3 Preliminary results

Dependent variable	Sales growth				Employment growth			
	Coef.	St. Error	Coef.	St. Error	Coef.	St. Error	Coef.	St. Error
Lagged size	-0.268 0.000	0.007	-0.269 0.000	0.007	-0.085 0.000	0.003	-0.085 0.000	0.003
Lagged cash holdings	-0.035 0.003	0.012	-0.033 0.005	0.012	0.046 0.000	0.007	0.047 0.000	0.007
Lagged leverage	-0.042 0.000	0.010	-0.040 0.000	0.010	-0.019 0.001	0.006	-0.018 0.001	0.006
Lagged profitability	-0.626 0.000	0.014	-0.626 0.000	0.014	0.173 0.000	0.007	0.173 0.000	0.007
Lagged asset tangibility	-0.073 0.000	0.007	-0.073 0.000	0.007	-0.003 0.384	0.004	-0.003 0.379	0.004
Lagged industry growth	0.157 0.000	0.013	0.159 0.000	0.013	0.035 0.006	0.013	0.034 0.006	0.013
Lagged size of competitors	-0.078 0.000	0.017	-0.076 0.000	0.017	-0.081 0.000	0.010	-0.080 0.000	0.010
Lagged industry competition	0.173 0.000	0.040	0.169 0.000	0.040	0.071 0.002	0.023	0.069 0.002	0.023
Lagged rule of law	0.094 0.000	0.020	0.092 0.000	0.020	0.035 0.011	0.014	0.035 0.012	0.014
Lagged GDP per capita	-0.112 0.042	0.055	-0.117 0.032	0.055	0.070 0.034	0.033	0.069 0.038	0.033
Lagged EPRI	0.055 0.000	0.008	0.024 0.005	0.009	0.032 0.000	0.005	0.022 0.000	0.005
Lagged payment duration	0.574 0.000	0.011	0.642 0.000	0.015	0.074 0.000	0.004	0.097 0.000	0.006
Lagged EPRI * lagged payment duration			0.128 0.000	0.015			0.043 0.000	0.007
Constant	3.624 0.000	0.569	3.662 0.000	0.568	0.361 0.296	0.345	0.368 0.286	0.345
Number of observations	605,212		605,212		582,960		582,960	
Firm- and year-fixed effects	YES		YES		YES		YES	
R squared	0.269		0.269		0.163		0.163	
F statistics	414.71	0.000	394.99	0.000	162.04	0.000	155.40	0.000

Standard errors are robust and clustered at the firm-level. P-values are reported below the coefficients.

root of the sum of the squared standard errors. Since medium firms are the least financially constrained firms in our sample, we interpret this result as indirect evidence that SMEs that have access to financial resources make use of these resources to grow by granting longer payment delays, an idea also present in Lefebvre (2021).

To gain further insights on the role of EPRI, we present in Table 5 the results of the firm- and year-fixed effects regressions on two distinct subsamples that correspond to countries where the EPRI increased over the considered period on to countries where the EPRI decreased. We consistently observe a positive and statistically significant relationship between payment duration and growth in employment and in sales in both subsamples. The interaction effect between payment duration and

Table 4 Firm- and year- fixed effects regressions of the relationship between payment duration and growth by size category

Dependent variable	Sales growth						Employment growth					
	Medium firms			Small firms			Medium firms			Small firms		
	Coef.	St. Error		Coef.	St. Error		Coef.	St. Error		Coef.	St. Error	
Lagged size	-0.474	0.039		-0.323	0.011		-0.306	0.009		-0.153	0.005	
	0.000			0.000			0.000			0.000		
Lagged cash holdings	0.206	0.058		0.036	0.017		-0.059	0.016		0.066	0.011	
	0.000			0.030			0.000			0.000		
Lagged leverage	0.052	0.047		-0.037	0.014		-0.045	0.013		0.006	0.009	
	0.264			0.009			0.001			0.511		
Lagged profitability	-0.540	0.070		-0.603	0.024		-0.662	0.017		0.135	0.013	
	0.000			0.000			0.000			0.000		
Lagged asset tangibility	-0.099	0.032		-0.034	0.010		-0.094	0.009		-0.002	0.006	
	0.002			0.000			0.000			0.750		
Lagged industry growth	-0.007	0.026		0.138	0.020		0.149	0.019		-0.009	0.018	
	0.787			0.000			0.000			0.638		
Lagged size of competitors	0.103	0.047		-0.092	0.022		-0.078	0.028		-0.064	0.014	
	0.029			0.000			0.005			0.000		
Lagged industry competition	0.157	0.151		0.139	0.060		0.200	0.052		0.035	0.035	
	0.300			0.021			0.000			0.293		
Lagged rule of law	0.138	0.051		0.089	0.026		0.122	0.035		0.018	0.017	
	0.007			0.001			0.001			0.284		
Lagged GDP per capita	-0.387	0.120		-0.095	0.068		-0.171	0.106		-0.039	0.089	
	0.001			0.163			0.106			0.661		
Lagged EPRI	0.012	0.021		0.024	0.011		0.021	0.015		-0.005	0.016	
	0.551			0.032			0.157			0.758		
Lagged payment duration	1.043	0.100		0.672	0.025		0.631	0.019		0.236	0.041	
	0.000			0.000			0.000			0.000		
Lagged EPRI * lagged payment duration	0.181	0.063		0.083	0.021		0.137	0.020		0.060	0.041	
										0.064	0.012	
										0.028	0.009	

EPRI is not statistically significant in countries where the EPRI increased. However, the interaction term is statistically significant and positive in countries where the EPRI decreased. In other words, the positive relationship between payment duration and growth is amplified by the quality of payment conditions, but only in countries characterized by relatively poor payment conditions. This observation corroborates hypothesis 2. We also compared the coefficients of the payment duration variable for the two subsamples. The difference, calculated as the difference between the coefficients over the square root of sum of the squared standard errors, is positive and statistically significant ($d=12.78$, $p<0.000$ for sales growth, $d=7.77$, $p<0.000$ for employment growth). This observation confirms that the relation between payment duration and growth is much stronger in countries where the EPRI increased. Overall, we find support for hypothesis 2.

Robustness tests

There are several important empirical issues in a study such as this one. More specifically, it is important to (1) assess whether our results really reflect the influence of trade credit practices on firm growth and not the other way around and (2) to examine whether our results are sensitive to other measures of payment duration. We thus conduct several robustness tests to assess the strength of our results.

The first issue we want to deal with is whether reverse causality concerns affect our data. As we already mentioned, it is possible that firms that have promising growth opportunities decide to extend more trade credit to facilitate growth. If this is the case, the positive relationship we observed in our regression results may simply reflect the fact that some firms have sustained high growth rates and that these firms decide to adjust their trade credit practices accordingly. To examine the existence of this problem, we use a propensity-score matching (PSM) approach to assess the impact of payment duration on growth (see Dehejia and Wahba (2002) for a detailed description of PSM). The general principle of a PSM method is to compare an outcome variable between a group of individuals (firms) that received a treatment to another group of matched individuals that did not receive the treatment. A difference in the outcome variable between both groups is attributable to the treatment if the individuals are correctly matched on a set of observable characteristics. In our case, the “treatment” is whether firms have above average payment durations and untreated (or control) firms are those that have below average payment durations.

Specifically, we create a dummy variable that equals to one if a firm’s payment duration is above its industry-country-mean payment duration and zero otherwise. We regress (via a logistic regression model) this dummy variable for the year 2011 (the beginning of our sample period) on the same set of variables as before: size, cash holdings, leverage, profitability, asset tangibility, industry growth, size of competitors, industry competition, GDP per capita, EPRI, and a set industry and country dummies. These regressors are the observable characteristics we use for matching. We derive from the results of this regression a propensity score that a firm has a payment duration above its industry-country-mean. This propensity score is the predicted probability of receiving treatment according to the logistic regression model.

Then, we match firms that have close propensities to have above industry-country mean payment duration and compare the likelihood to have a growth rate above the industry-country average for treated firms (those which have above industry-country mean payment duration) and control firms (those which have below industry-country mean payment duration). We use the command “psmatch 2” available for Stata 16 to perform the matching. The results of the PSM procedure (selection equation, average treatment effect, and post-hoc balancing properties tests) are unreported for brevity but available upon request. These results confirm that firms that have above industry-country mean payment duration have a greater likelihood to experience a growth rate above the industry-country mean. In other words, the PSM procedure confirms that payment duration impacts growth.

We conduct a second set of robustness checks to ensure that our results are not sensitive to the choice of the payment duration measure used. In the previous estimations, we use the receivables on sales ratio as an independent variable. This measure of payment duration does not take into account industry- and country-specificities of trade credit. For instance, some countries offer derogatory legal payment delays for some industries, which means that a given payment duration can be legal in one country but not in another. In addition, trade credit practices and norms vary from one industry to another, which means that a payment duration can be above the standard in an industry but below the standard in a different industry. Variations in macro-economic factors such as commodities prices and interest rates differently impact industries leading to heterogenous changes in trade credit practices across industries (Ernest & Young, 2019).

Building on previous research on trade credit and working capital management (Aktas et al., 2015; McGuinness & Hogan, 2016), we use two alternative payment duration measures. The first alternative measure of payment duration is the country-industry-mean-adjusted receivables on sales ratio. Concretely, we calculate for each country, industry, and year, the mean of the receivables on sales ratio. Then, for each firm-year observation, we subtract the mean receivables on sales ratio to a firm’s receivables on sales ratio and use the result as an independent variable. Doing so ensures that we capture deviations to credit duration standards. The results, unreported for brevity but available upon request, are similar to those obtained in Table 5.

The other alternative measure of payment duration we use is based on a two-stage regression approach (or “predicted value” approach) as in Aktas et al. (2015). Specifically, we first regress the receivables on sales ratio on a set of factors known to influence it, namely firm size, cash holding, leverage, and market share (sales turnover divided by the country industry annual total volume of sales). These regressions are performed separately for each country, each industry, and each year separately. Then, the residuals of these first-stage regressions are used as the independent variable in our main estimations. This approach allows us to control for the endogenous determination of the amount of trade credit granted to customers by firms’ managers and account for the bargaining power SMEs have via the market share variable. The results, unreported for brevity but available upon request are fully comparable to those obtained with the standard receivables on ratio. Taken together, the results obtained with these two alternative measures of payment duration bring additional credibility to our findings.

Table 5 Comparison between countries where the EPRI increased and countries where the EPRI decreased

Dependent variable	Countries where the EPRI increased				Countries where the EPRI decreased			
	Sales growth		Employment growth		Sales growth		Employment growth	
	Coef.	St. Error	Coef.	St. Error	Coef.	St. Error	Coef.	St. Error
Lagged size	-0.289	0.009	-0.091	0.005	-0.275	0.008	-0.086	0.003
	0.000		0.000		0.000		0.000	
Lagged cash holdings	0.108	0.019	0.042	0.011	-0.060	0.015	0.054	0.009
	0.000		0.000		0.000		0.000	
Lagged leverage	0.236	0.016	0.007	0.010	-0.100	0.012	-0.020	0.007
	0.000		0.490		0.000		0.002	
Lagged profitability	-0.483	0.021	0.221	0.013	-0.665	0.017	0.152	0.009
	0.000		0.000		0.000		0.000	
Lagged asset tangibility	-0.048	0.011	0.000	0.007	-0.077	0.008	-0.002	0.004
	0.000		0.984		0.000		0.643	
Lagged industry growth	0.046	0.016	0.056	0.021	0.077	0.020	-0.041	0.016
	0.004		0.009		0.000		0.009	
Lagged size of competitors	-0.136	0.025	-0.094	0.015	0.051	0.023	-0.020	0.013
	0.000		0.000		0.030		0.111	
Lagged industry competition	0.058	0.065	0.041	0.036	0.205	0.050	0.078	0.029
	0.376		0.260		0.000		0.008	
Lagged rule of law	0.231	0.037	0.145	0.030	0.565	0.081	0.094	0.040
	0.000		0.000		0.000		0.018	
Lagged GDP per capita	-1.158	0.113	-0.691	0.077	-1.055	0.112	-0.359	0.060
	0.000		0.000		0.000		0.000	
Lagged EPRI	0.021	0.016	-0.007	0.010	-0.008	0.013	-0.007	0.007
	0.191		0.447		0.529		0.339	
Lagged payment duration	1.044	0.034	0.188	0.013	0.562	0.017	0.076	0.007
	0.000		0.000		0.000		0.000	
Lagged EPRI * lagged payment duration	-0.045	0.036	-0.003	0.016	0.123	0.017	0.042	0.008

Table 5 (continued)

Dependent variable	Countries where the EPRI increased			Countries where the EPRI decreased		
	Sales growth		Employment growth	Sales growth		Employment growth
	Coef.	St. Error	Coef.	Coef.	St. Error	Coef.
Constant	0.212		0.866	0.000		0.000
	14.539	1.176	8.230	12.234	1.136	4.435
	0.000		0.000	0.000		0.000
Number of observations	207,184		202,218	398,028		380,742
Firm- and year-fixed effects	YES		YES	YES		YES
R squared	0.296		0.201	0.267		0.151
F statistics	237.82	0.000	117.36	246.15	0.000	84.55

Standard errors are robust and clustered at the firm-level. p-values are reported below the coefficients.

Conclusion

Discussion

Our main goal in this paper was to study the role of payment duration and institutional payment conditions on firm growth. There are conflicting views about the role of payment duration on growth, since the prevailing academic perspective considers payment duration as a way to reduce information asymmetries that should facilitate growth, while practitioners and public policies stress the detrimental impact of late payment and uncertainty associated with long payment duration on growth. Our results bring empirical evidence of a positive relationship between payment duration and growth for European firms and especially SMEs, suggesting that granting trade credit to customers is a channel to growth. These findings corroborate theoretical frameworks (Smith, 1987; Long et al., 1993) and previous empirical investigations of the role of trade credit on SMEs performance (García-Teruel & Martínez-Solano, 2010; Martínez-Sola et al., 2013) and extend them to a vast international sample.

Furthermore, we build on institutional reforms and the integration of the LPD by EU countries that were designed to improve payment conditions. The heterogeneity in the way the LPD was integrated in EU countries allowed us to identify countries that experienced a strong improvement of legal payment conditions and to use the change in legal payment conditions in these countries as an exogenous factor. Our results show that the positive relationship between payment duration and growth is amplified by the change in legal payment conditions, confirming the relevance of a strong implementation of the LPD in EU countries, especially in those countries where payment conditions are relatively poor.

An important question remains, however, unanswered. To what extent are SMEs' managers aware of the achievement of growth through trade credit granted to customers and did they deliberately choose to grow via this channel? A common argument views the extension of trade credit as the result of negotiations between customers and suppliers, where the larger or dominant player imposes payment conditions (Mian & Smith 1992; Ng et al., 1999). According to this view, the extension of trade credit by SMEs is not a choice but the "logical" outcome of business relationships with larger, more powerful firms. However, it seems important to acknowledge that, while large firms are potentially important customers for SMEs, they do not necessarily account for all their customers, neither do they account for the majority of their sales turnover. SMEs have an incentive to grant trade credit to other SMEs if these SMEs are financially constrained, because doing so means that the supplier acts as a substitute to other financiers, like banks (García-Teruel & Martínez-Solano, 2010; McGuinness & Hogan, 2016). An obvious limitation of the practice of granting trade credit to financially constrained customers is the increased risk of non-payment, should customers eventually make default (Ng et al., 1999).

However, these risks are likely counterbalanced by the benefits obtained by offering financial stability through the extension of trade credit. For an SME, strengthening business relationships with other SMEs through trade credit creates a network of "captive" customers if these other SMEs have limited access to finance (Shipley & Davies, 1991). In other words, SMEs that grant trade credit facilitate the growth of

their customers, which in turn feeds the SME's growth, creating a virtuous circle. Our results are consistent with this view and further highlight that improvements in legal payment conditions contribute to enhance the benefits of granting trade credit by ensuring that, should payments be excessively delayed, SMEs would have enforcement tools at hand.

From a theoretical perspective, it is interesting to notice that our work fits rather well in ongoing discussions about the role of institutional changes as “external enablers” of entrepreneurial activity (Tolbert et al., 2011; Kimjeon & Davidsson, 2022). For instance, Eberhart et al. (2017) document that the adoption of less stringent bankruptcy laws in Japan led to an increase in the founding of high-growth ventures. We show that changes in the institutional environment, like the LPD, are designed to (European Commission, 2015) and do facilitate growth, confirming the central role of public policies in supporting entrepreneurship. The adoption of the LPD in the EU is a clear effort to limit late payment risk for SMEs, and it has largely been acknowledged that reducing uncertainty and risk is critical in supporting SMEs (Scott, 2008). Whether these benefits are serendipitous or actively sought by SMEs' managers remains an interesting and open question that future research could address, for instance via surveys.

Practical implications

Our work has important implications for both policy makers and SMEs' managers. The negative effect on growth observed in countries where the EPRI decreased highlights the need to strengthen legal payment conditions, especially in countries where late payment is already an issue. The benefits in terms of growth employment we observe in countries that experience an increase in EPRI show how fruitful it is to actively implement the LPD. Furthermore, managers and entrepreneurs should carefully consider payment conditions when choosing where to start a new activity, since institutional payment conditions have a strong influence on growth.

It has been well documented that active financial management is an important but often neglected aspect of strategic management for SMEs desiring to achieve growth (Peel & Wilson, 1998; Peel et al., 2001; Brinckmann et al., 2011). Our results show that when SMEs grant trade credit, they grow faster. Previous research has documented the benefits of granting trade credit for SMEs' profitability (Martínez-Sola et al., 2013) and we extend them to SMEs' growth. It seems therefore useful for SMEs' managers to dedicate more time to financial management or to bring additional financial skills to help integrate trade credit as a full dimension of strategic planning.

Limitations

There are several empirical issues regarding the estimation procedure and the variables used in a study such as this one. The results of the regression analysis may be affected by endogeneity, a concern that is in part alleviated by the use of the PSM procedure. The accounts receivable on sales ratio does not take into account industry or country specificities, neither does it reflect whether a firm decides to grant trade credit to customers or is constrained to extend trade credit by customers that have

strong bargaining power. We thus use alternative measures of the extension of trade credit that take into account country- and industry-standards as well as a two-stage regression approach to measure deviations from such standards. These efforts bring credibility to our results.

Despite these efforts, we see several limitations to our work. First, we were not able to identify the structure of customers for our sample firms due to a limitation of our database. Depending on the number of customers and their relative importance in sales volume, firms may well adopt differentiated trade credit practices (Wu et al., 2020). For instance, it is possible that firms deliberately decide to grant more trade credit to financially constrained customers only in order to improve the financial health of the entire supply chain (Hofmann & Kotzab, 2010). Future research could adopt such a fine-grained analysis, although we see the data collection process as quite challenging.

Second, it seems important to acknowledge that our results may not be generalizable to other countries. When compared to other geographic areas, the EU remains a privileged context in terms of banking development and institutional settings. The example of the implementation of the LPD shows that public policies are designed to reduce late payments and improve trade credit conditions. At the same time, when new countries join the EU, the need to incorporate relatively fast new institutional standards in terms of payment delay can be especially challenging for SMEs growth (Smallbone & Rogut, 2005). Regular practitioners' reports regularly highlight improvements in trade credit conditions and working capital management in the EU and the US while such improvements are less clear in other countries (Ernest & Young, 2019). There is thus considerable heterogeneity in the strength of the institutional context for payment conditions, so our observations of a positive relationship between the extension of trade credit and growth may not be generalizable to other countries. More empirical research is thus required to examine whether and to what extent the extension of trade credit by SMEs has a positive influence on growth in other institutional settings.

The previous limitation is all the more important as SMEs' financing patterns around the world are highly heterogeneous (Beck et al., 2008). One of our main theoretical arguments was to view the extension of trade credit by SMEs as a channel to build strong trade relationships with customers. This argument implicitly assumes that trade credit extension by SMEs is a viable alternative (or complement) to bank financing, which may not be true to the same extent in other countries (Fisman & Love, 2003; Machokoto et al., 2022). In the context of our study (EU), we considered countries that have bank-oriented economies, meaning that banks and formal credit institutions are well developed. It seems reasonable to believe that the extension of trade credit by SMEs and the benefits obtained in terms of growth strongly depend on the degree of development of the banking sector. For instance, it has been shown that a low development of financial institutions is associated with a stronger use of trade credit for firm growth (Ge & Qiu, 2007). Future research could thus fruitfully examine the imbrication of legal payment conditions, financial development, and the role of trade credit for SMEs growth.

Concluding remark

For sure, the Covid-19 crisis will have a strong impact on SMEs' growth and survival via the trade credit channel since customers' bankruptcies endanger suppliers. However, trade credit may not only be the source of a contagion effect but also a resilience factor for the SMEs' sector. Extending payment delays in times of crisis may help distressed firms overcome liquidity shortages, which in turn increases the likelihood of customers' survival. Could trade credit have such a cushioning effect? Hopefully, our work will bring food for thought to SMEs' managers and public policies about the role of trade credit and payment duration.

Appendix

Summary statistics by country

Country	Number of observations	EPRI 2010	EPRI 2017	Median sales growth	Median employment growth	Median payment duration
Germany	733	-0.060	0.030	0.027	0.009	0.059
Austria	123	-0.060	0.490	0.044	0.000	0.073
Belgium	12,627	-0.120	-0.110	0.017	0.000	0.135
Spain	179,183	-0.340	-0.040	0.029	0.000	0.196
Finland	28	0.480	0.240	0.028	0.000	0.069
France	13,261	0.000	0.120	0.023	0.000	0.148
Greece	5,661	-0.600	-0.660	0.023	0.000	0.286
Italy	363,115	-0.260	-0.560	0.010	0.000	0.182
Netherlands	7	-0.060	0.090	0.020	0.000	0.128
Portugal	34	-0.700	-1.080	-0.018	0.000	0.134
Sweden	25,619	0.400	0.090	0.011	0.000	0.084
Total sample	600,431	-0.060	0.030	0.016	0.000	0.177

The median value over the total sample is reported in the last row of the table for each variable.

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Declarations

Conflict of interest The author has no conflict of interest to report.

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